

ECE 210 Midterm 3 Worksheet

Note: This worksheet is not guaranteed to be entirely representative of the midterm's contents. Material may appear on this worksheet which will not appear on the midterm, and vice versa.

Partial Fraction Expansion

1) (Technique 1)

a) sub problem 1

Solution sub problem 1 solution

$$sF(s) = s \frac{s+3}{s(s+2)(s+5)} = \dots$$

b) sub problem 2

Solution sub problem 2 solution

2) (Technique 2)

a) sub problem 1

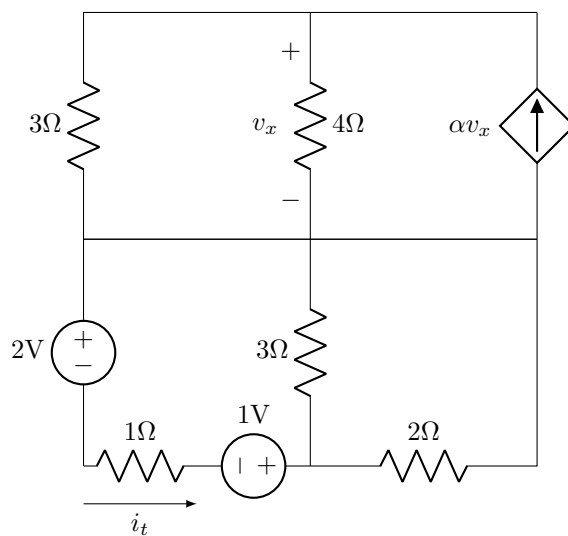
Solution sub problem 1 solution

b) sub problem 2

Solution sub problem 2 solution

S-Domain Circuit Analysis

3) Consider the circuit below (SAMPLE CIRCUIT - CHANGE).



a) sub problem 1

Solution sub problem 1 solution

b) sub problem 2

Solution sub problem 2 solution

Marginal Stability & Resonance

4) problem 1

a) marginal stability part

Solution sub problem 1 solution

b) resonance part

Solution sub problem 2 solution

c) etc

Solution sub problem 3 solution

5) problem 2

a) marginal stability part

Solution sub problem 1 solution

b) resonance part

Solution sub problem 2 solution

c) etc

Solution sub problem 3 solution